5. Newton - Raphson method w/ $x_0 = 0$

General eqn: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

From given eqn: $x^3 - 7x^2 + 14x - 5$

$f'(x) = 3x^2 - 14x + 14 = 0$

$$x_{n+1} = x_n - \frac{[x_n^3 - 7x_n^2 + 14x_n - 5]}{[3x_n^2 - 14x_n + 14]}$$

$$x_1 = x_0 - \frac{[x_0^3 - 7x_0^2 + 14x - 5]}{[3x_0^2 - 14x_0 + 14]}$$

$$= 0 - \frac{0 - 0 + 0 - 5}{0 - 0 + 14} = \frac{5}{14}$$

$f(x_1) = \left(\frac{5}{14}\right)^3 - 7\left(\frac{5}{14}\right)^2 + 14\left(\frac{5}{14}\right) - 5 = -0.8473$

$$x_2 = \frac{5}{14} - \frac{\left(\frac{5}{14}\right)^3 - 7\left(\frac{5}{14}\right)^2 + 14\left(\frac{5}{14}\right) - 5}{3\left(\frac{5}{14}\right)^2 - 14\left(\frac{5}{14}\right) + 14} = \frac{1920}{4291}$$

$f(x_2) = \left(\frac{1920}{4291}\right)^3 - 7\left(\frac{1920}{4291}\right)^2 + 14\left(\frac{1920}{4291}\right) - 5 = -0.0476$

This pattern is continued for each new iteration. I will just show what the evaluations are equal to.

$$x_3 = \frac{1920}{4291} - \frac{f(1920/4291)}{f'\left(\frac{1920}{4291}\right)} = \frac{2984714500053}{6586455603 14} = 0.453$$

$f(x_3) = -1.84 \times 10^{-4}$

$x_4 = A$ by function $= 0.4532$

| $x_n$ | x (frac) | x (decimal) | $f(x_n)$ |
|---|---|---|---|
| $x_0$ | 0 | 0 | -5 |
| $x_1$ | 5/14 | 0.3571 | -0.8473 |
| $x_2$ | 1920/4291 | 0.4474 | -0.0476 |
| $x_3$ | 2984714500053 / 6586455603 14 | 0.45316 | $-1.84 \times 10^{-4}$ |
| $x_4$ | By fnct | 0.4532 | $-2.8 \times 10^{-9}$ |